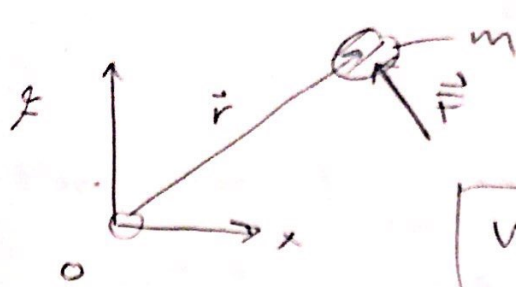


Work, power, + energy



LMB: $\sum \vec{F} = m\vec{a}$

$\int \vec{v} \cdot \{ \} dt$

$W_{12} = \int_1^2 P dt = E_{k2} - E_{k1}$

Always for all forces

Add to ODEs: $\dot{W} = \vec{F}(\vec{r}, \vec{v}, t) \cdot \vec{v}$

Special forces: conservative forces

I. $\vec{F}(\vec{r}, \vec{v}, t) = \vec{F}(\vec{r})$

↑ only depends on position

II. One of these special properties (one implies all)

I. Work is path independent

$\int_{\vec{r}_1}^{\vec{r}_2} \vec{F}(\vec{r}) \cdot d\vec{r}$ is independent of path
for all paths in region

Example:

$\vec{F} = -C \frac{\vec{r}}{|\vec{r}|^3}$

inverse square gravity

coulomb's law

$\vec{F} = -\frac{\vec{r}}{|\vec{r}|} k(|\vec{r}| - l_0)$ Hooke's law

~~Gradient of the force is zero $\nabla \vec{F} = \vec{0}$ x~~

II. $\vec{F} = -\vec{\nabla} E_p(\vec{r})$

↑ potential energy

III. $E_k + E_p = 0$

IV: $\vec{\nabla} \times \vec{F} = 0$

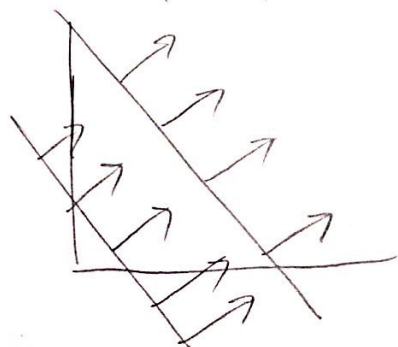
$$\frac{dF_x}{dy} = \frac{dF_y}{dx}$$

More famous examples:

ex) near earth gravity: $\vec{F} = -mg\hat{j}$

$$E_p = mgy$$

ex) Any force in a const dir w/ magnitude only depend on distance in that direction



$$E_p = -\int_0^s f(s') ds'$$

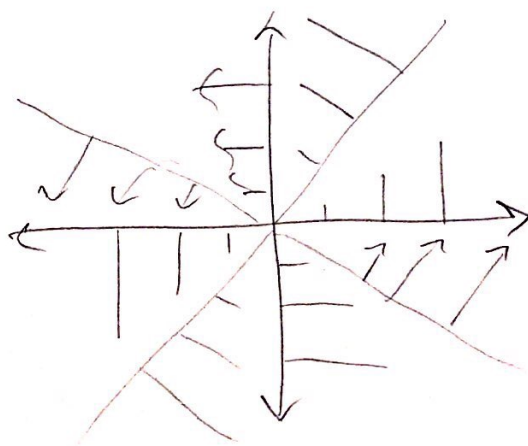
$$dW = d\vec{r} \cdot \hat{\lambda} f(s)$$

Counter ex)

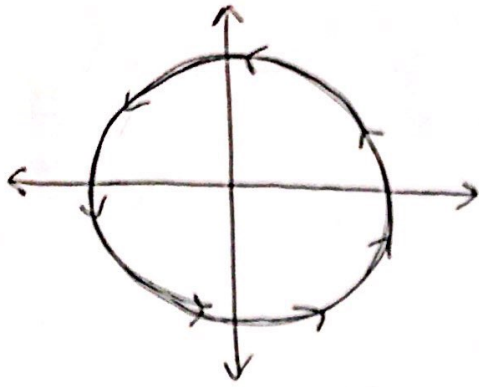
$$F_x = -y$$

$$F_y = x$$

$$\vec{F} = \hat{k} \times \vec{r}$$



Conservative Force Field



$$(1) \text{ curl: } \vec{\nabla} \times \vec{F} = \frac{dF_y}{dx} \hat{k} - \frac{dF_x}{dy} \hat{k} \\ = \hat{k} + \hat{k} = 2\hat{k} \neq 0$$

$$\frac{dF_x}{dy} \stackrel{?}{=} \frac{dF_y}{dx}$$

$-1 \neq 1 \Rightarrow$ Not a conservative force field

$$(2) \oint dW > 0$$

$\neq 0 \Rightarrow$ Not a conservative force field

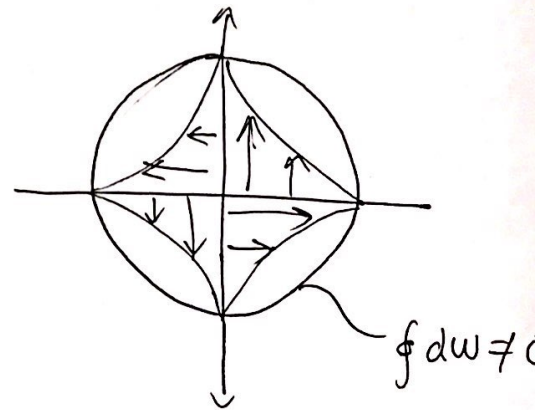
(3) Is there a E_p s.c

$$-\frac{dE_p}{dx} = -y \quad ; \quad -\frac{dE_p}{dy} = x \quad (\text{No!})$$

$$\vec{F} = \frac{\hat{k} \times \vec{r}}{|\vec{r}|^2}$$

$$F_y = -\frac{x}{x^2+y^2}, \quad F_x = \frac{y}{x^2+y^2}$$

$$\frac{dF_x}{dy} \neq \frac{dF_y}{dx}$$

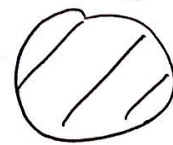


Note: E_p is an additive constant
 $\rightarrow$ still good

$$\underbrace{E_p}_{\text{if exists}} + E_k = \text{constant}$$

$$W_{12} = -\Delta E_p = \Delta E_k$$

For this region



Not conservative



conservative